Flight Controls – General Data

1. General

The flight control surfaces are operated in the conventional manner by cables from a dual control column and dual rudder pedals, allowing the aircraft to be flown from either the pilot's or co-pilot's position. The ailerons, elevators and rudder are conventional, but the ailerons are hinged to arms at the trailing edge of each outboard fore flap. The elevators and rudders are aerodynamically horn balanced and, together with the ailerons, are internally mass balanced. Geared tabs are fitted to the ailerons and rudder to provide an aerodynamic assist to control surface movement. Trim tabs, which are adjustable in flight, are fitted to the left aileron, left elevator, and rudder. The right elevator incorporates a trim tab which is interconnected with the wing flap system, and trims the elevators in proportion to wing flap movement. Each cable circuit is provided with conveniently located turnbuckles, and push-pull rods which are adjustable by means of screw threaded end adapters. The wing flaps are interconnected by push-pull rods which are operated by a single hydraulically-operated actuator mounted in the cabin roof.

2. Servicing

CAUTION **IF REMOVING OR INSTALLING CONTROL CABLES, DISCONNECT MAIN BATTERY (REFER TO 24-30-31, MAIN BATTERY SYSTEM – MAINTENANCE PRACTICES, PARA 2.), AVIONICS START BATTERY (IF INSTALLED, REFER TO 24-30-32, AVIONICS START BATTERY (MOD 6/2315) – GENERAL DATA, PARA 4.) AND EMERGENCY BATTERY (IF INSTALLED, REFER TO 33-50-01, EMERGENCY LIGHTING BATTERY – GENERAL DATA, PARA 1.) AND DISCONNECT GROUND POWER.**

A. General Rigging Instructions

The rigging of each control system of the flight controls is detailed in the Maintenance Practices of the individual systems. For each procedure it is assumed that all components are correctly installed, that the cables are fitted but not tensioned, and that all jam nuts on push-pull rods and turnbuckles are locked on the completion of rigging. It is also assumed that access panels and upholstery are removed, as required, for accomplishing these procedures and are replaced on completion of the operation. Figure 1 in 12-00-00, Servicing – General Data, provides locations of access and inspection provisions. The hydraulic system must be fully serviceable before wing flap, aileron, or flap/elevator interconnect trim rigging is started.

B. Tensioning of Control Cables

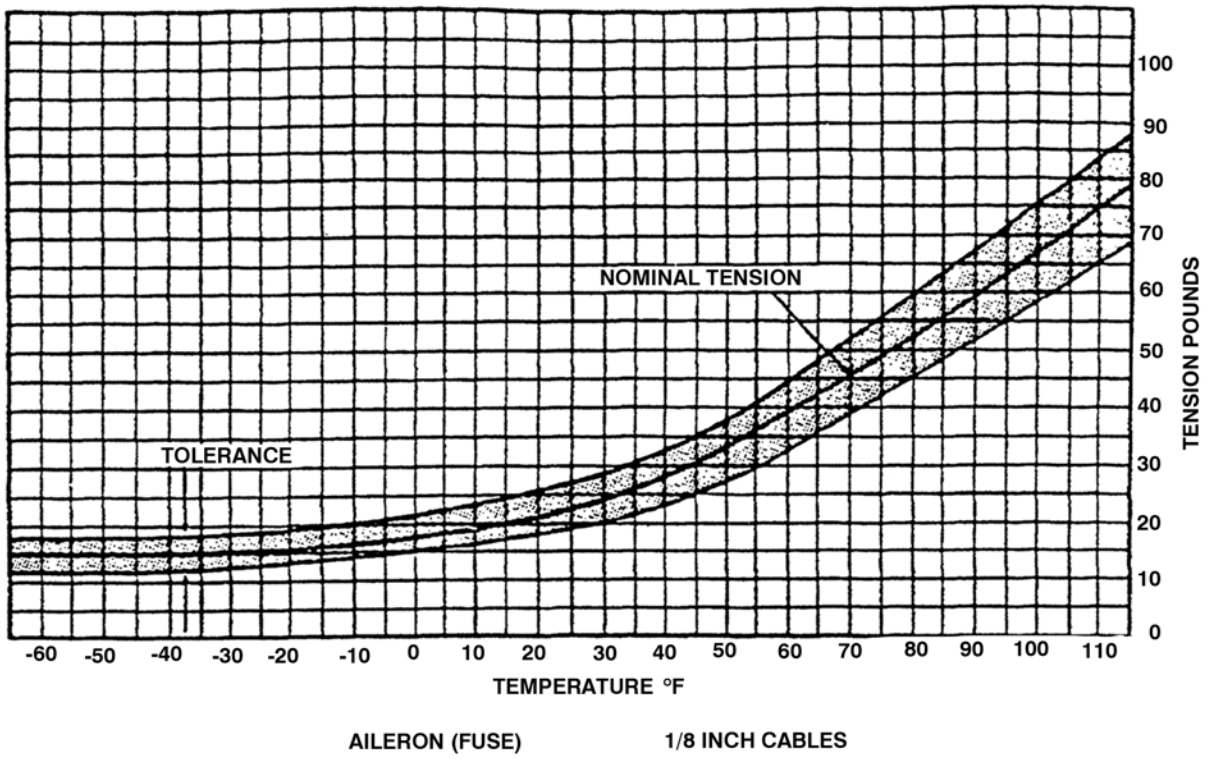
Whenever flight control system cables are slackened or replaced, they must be retensioned to the value shown on the appropriate graph (refer to Figure 1). Where possible, tensiometer readings are to be taken on bare cable. For cables fully covered in nylon, nominal values 15% lower than those shown on Figure 1 are to be used. When checking cable tensions, observe the following precautions:

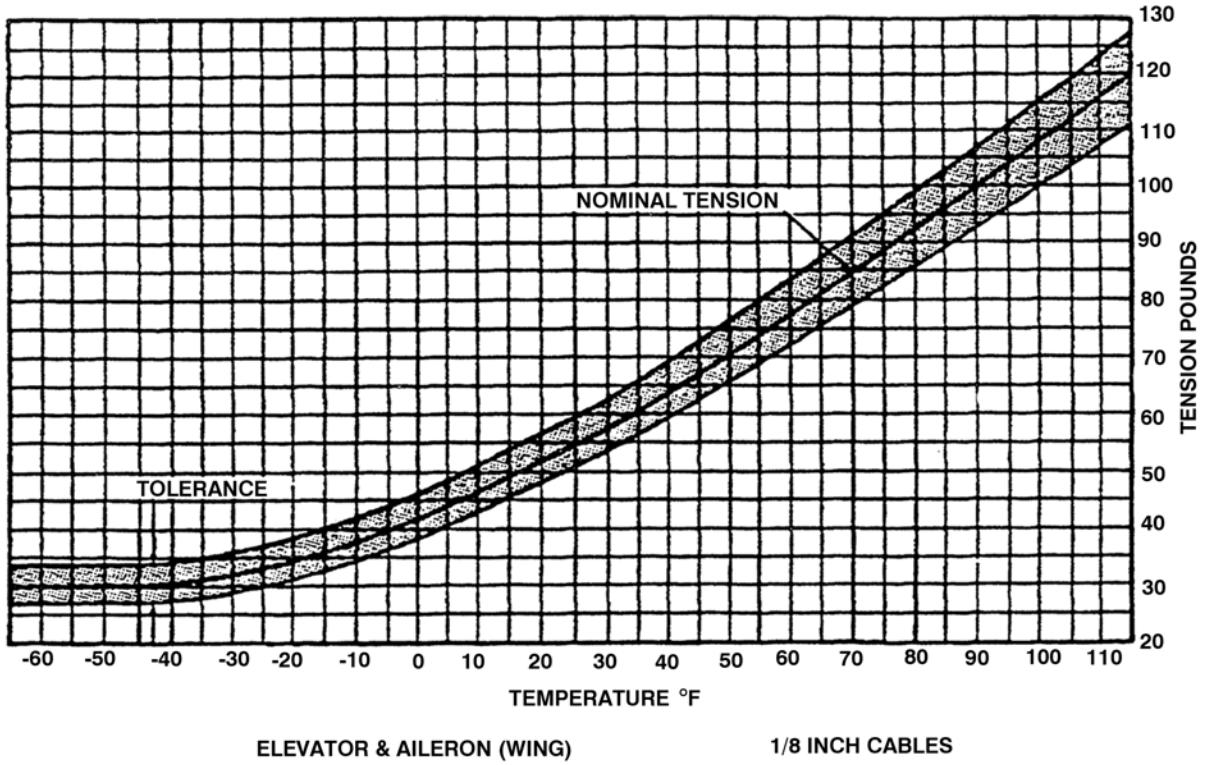
1) Ensure that tensiometer is of correct type for cable.

2) Vibrate cables to prevent an incorrect reading due to friction in circuit.

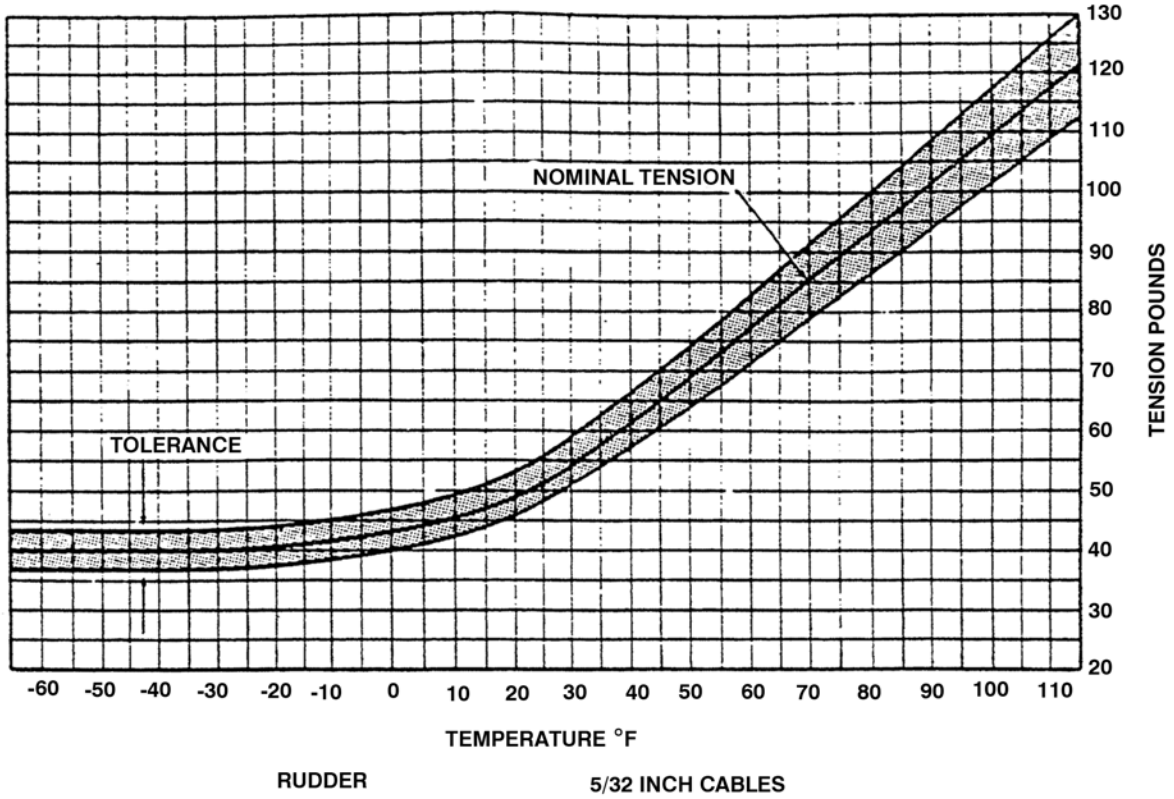
3) Position tensiometer on center portion of cable and not close to a quadrant, bellcrank, turnbuckle, etc.

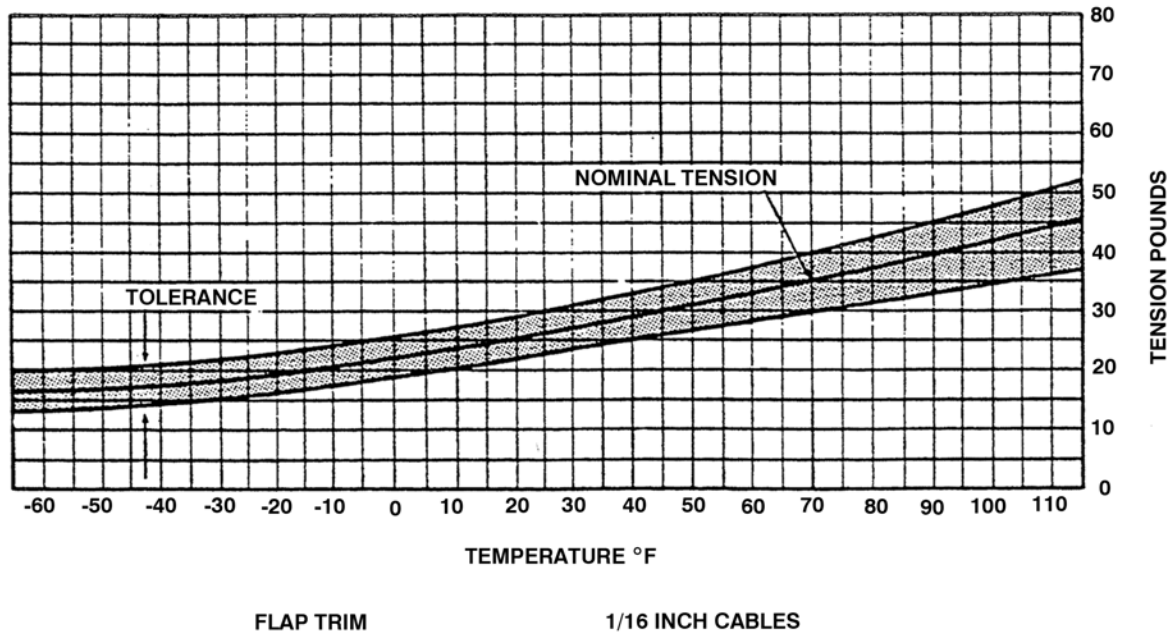
4) Recheck cable tensions after locking turnbuckles.



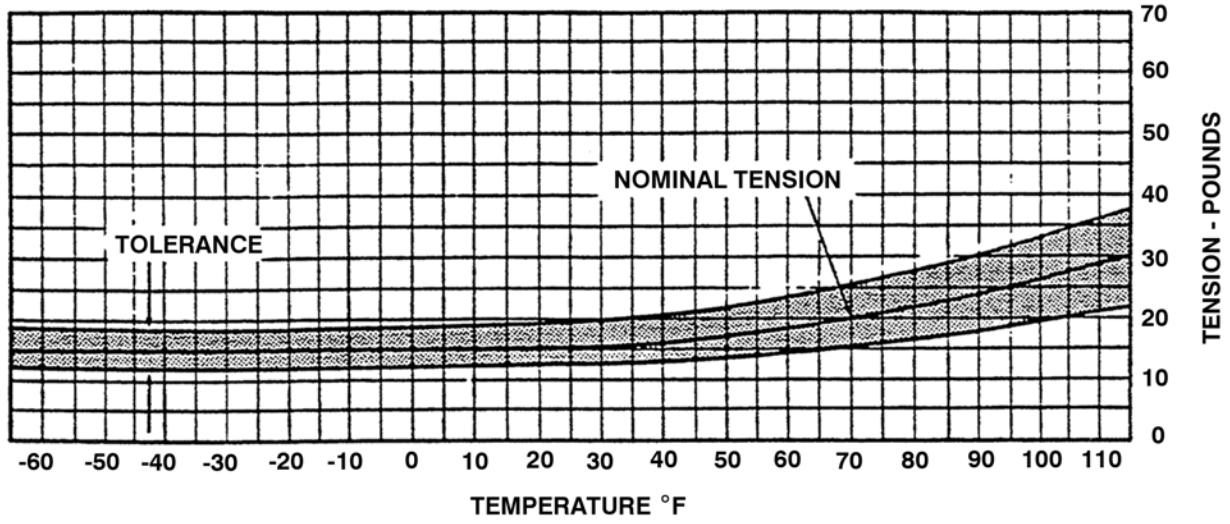


Cable Tension Graphs
Figure 1 (Sheet 1 of 3)





Cable Tension Graphs
Figure 1 (Sheet 2 of 3)



ELEVATOR TRIM
RUDDER TRIM
FLAP FOLLOW-UP
FLAP SELECTOR

1/16 INCH CABLES

ON AIRCRAFT FITTED WITH AN AUTOPILOT AUTOMATIC PITCH TRIM ACTUATOR, READ NOMINAL TO MAX. TENSION FOR ELEVATOR TRIM e.g. TENSION AT 70°F IS 20^{+5}_{0} POUNDS.

Cable Tension Graphs
Figure 1 (Sheet 3 of 3)

3. Inspection/Check

CAUTION **IF REMOVING OR INSTALLING CONTROL CABLES, DISCONNECT MAIN BATTERY (REFER TO 24-30-31, MAIN BATTERY SYSTEM – MAINTENANCE PRACTICES, PARA 2.), AVIONICS START BATTERY (IF INSTALLED, REFER TO 24-30-32, AVIONICS START BATTERY (MOD 6/2315) – GENERAL DATA, PARA 4.) AND EMERGENCY BATTERY (IF INSTALLED, REFER TO 33-50-01, EMERGENCY LIGHTING BATTERY – GENERAL DATA, PARA 1.) AND DISCONNECT GROUND POWER.**

A. Inspection of Control Cables

Inspect control cables at prescribed inspection intervals to ensure their serviceability. Before inspecting cables installed in an aircraft, ensure that all accessible portions of the cable run are clean. Dust and metal particles will readily adhere to grease and impair smooth operation of the controls. Cables in the cabin area of fuselage, though not accessible along their entire run, can be cleaned and inspected at access points where they pass over pulleys or through fairleads where broken wires and wear are most likely to occur. Move flight controls through their full range of travel to allow for maximum cable length inspection. A control tending to foul momentarily is often an indication of a frayed cable fouling on a fairlead or pulley. A simple method of detecting broken wires is by running a lint-free rag along the cable. The rag will snag on any projections. If worn, frayed or corroded cables are suspected, slacken turnbuckles to allow careful twisting of cables to open strands for a more detailed inspection. Replace cables if worn or corroded and if wear limits in 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data are exceeded. Always replace cables where serviceability is marginal.

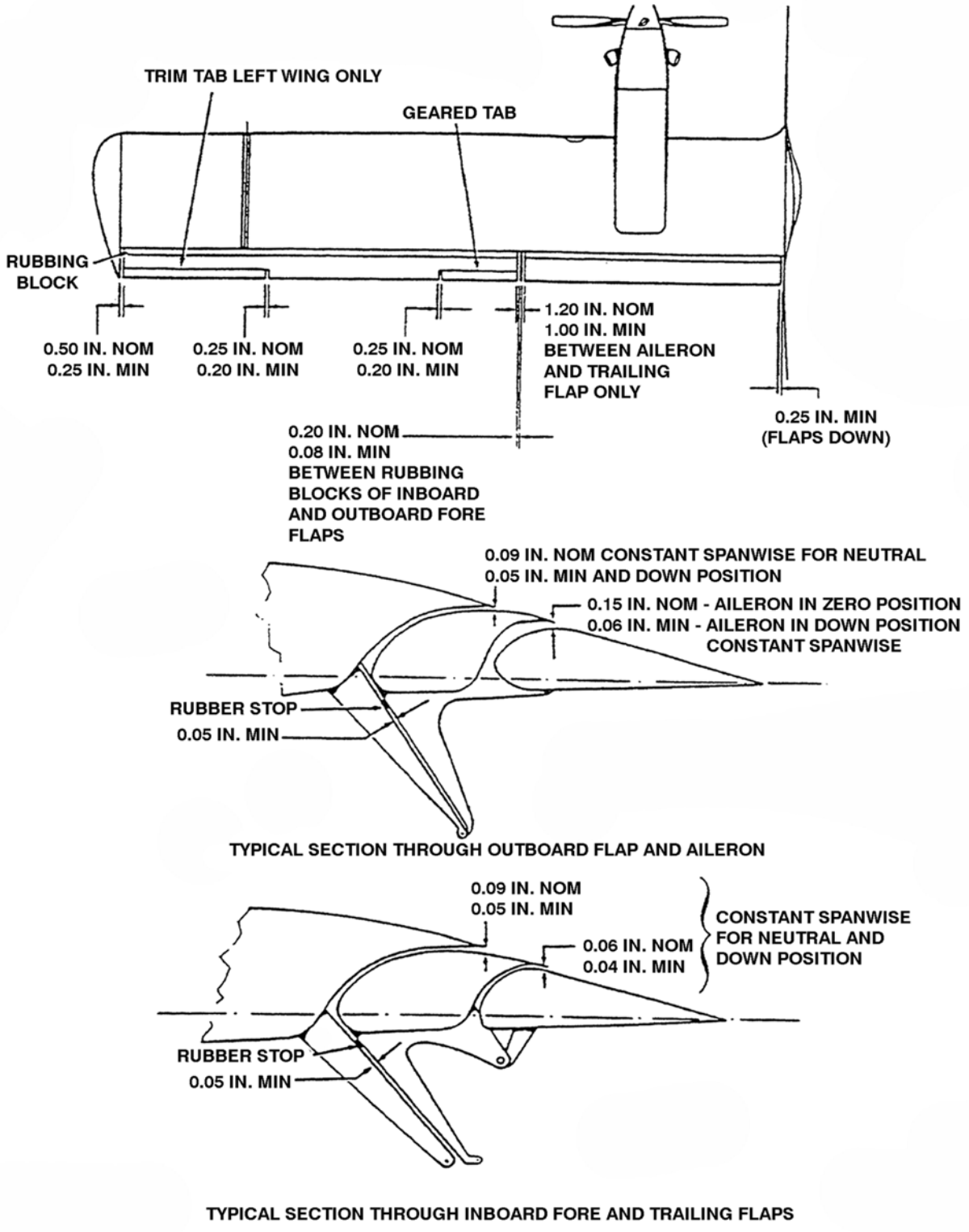
Remove corrosion and or zinc dust from the external surfaces in accordance with 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data.

B. Checking Control Surface Clearances

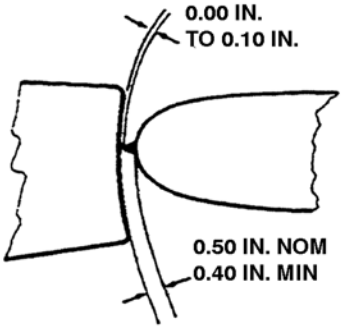
To check individual control surface clearances, refer to dimensions given in Figure 2.

C. Checking Control Components

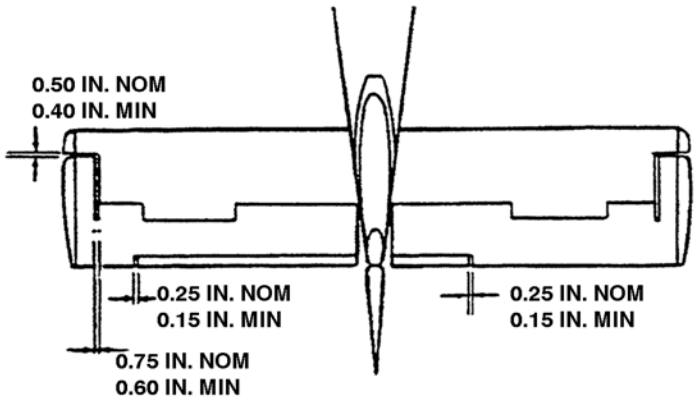
Inspect all control surfaces for damage and freedom of movement. Inspect individual control components for security. Refer to Inspection Requirements Manual for periodic and internal inspection requirements.

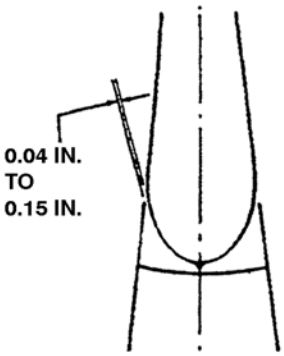


Control Surface Clearances
Figure 2 (Sheet 1 of 2)

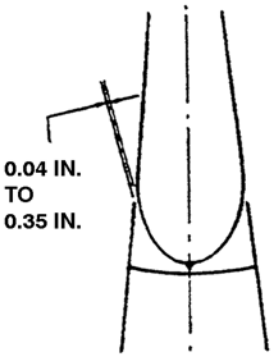


TYPICAL SELECTION
THROUGH ELEVATOR

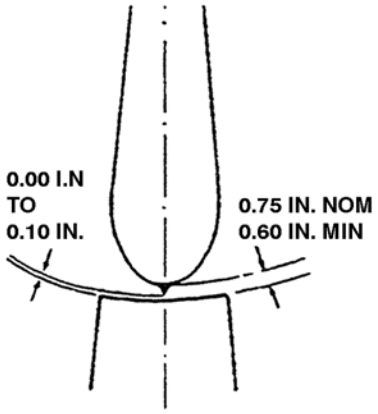




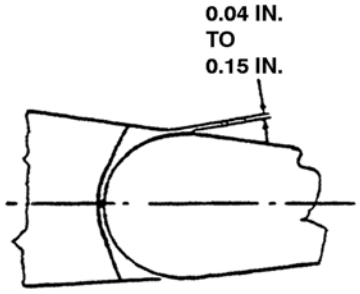
RUDDER/SHROUD CLEARANCE



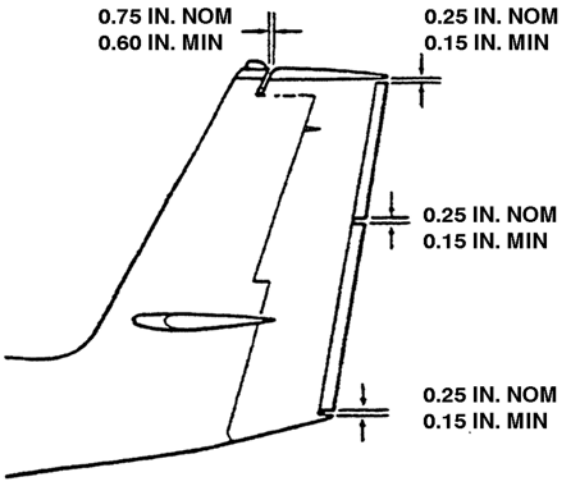
RUDDER/FUSELAGE
FAIRING CLEARANCE



TYPICAL SECTION
THROUGH RUDDER



ELEVATOR/SHROUD CLEARANCE



Control Surface Clearances
Figure 2 (Sheet 2 of 2)

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